

LARGE LANGUAGE MODELS IN HUMAN-MACHINE INTERACTION

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Abstract

This document serves as a template for the seminar write-up on Large Language Models in human-machine interaction. It will be replaced with the final abstract once the paper is complete.

Keywords: Large Language Models, Human-Machine Interaction, Seminar

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1 Introduction

In high-stakes environments such as aerospace, medicine, disaster response, military applications, and similar domains, the effectiveness and speed with which operators interact with the tools available to them are of critical importance. Traditionally, human-machine interaction (HMI) in such domains has been achieved through methods that have been field-tested over many years, such as user interfaces (UIs) utilizing buttons, joysticks, haptic-feedback sensors, and touchscreens. Depending on the application domain, specific norms and regulations govern the permissible interaction mediums for HMI user interfaces, including standards such as IEC 60204-1 for industrial electrical equipment and NASA-STD-3001 for aeronautical systems.[Int16][CWF23]

However, with the introduction of transformer encoder-decoder models and the subsequent popularization of large language models such as Google's BERT, OpenAI's GPT-3, and more recently GPT-5, a wide range of new possibilities has emerged in the field. Moreover, recent advancements in large language models in the areas of speech-to-speech and visual language models have created additional and particularly interesting opportunities.

This paper presents a literature review of recent advancements in relation to human-machine interaction within the context of safety-critical working environments.

2 Related Work

2.1 Text-Based LLM Interfaces

Since the introduction of the Transformer architecture in "Attention Is All You Need", text-based content generation and processing have become the central foundation of modern natural language processing (NLP).[VSP⁺17] In this context, large language models are a family of neural language models trained on vast amounts of textual data and capable of performing a wide range of natural language processing tasks, such as generating context-aware responses based on user-provided text prompts.[BMR⁺20] To utilize this family of models as a user interface, the model must be capable of understanding domain-specific context and of acting upon user prompts, rather than merely generating textual responses. Several methods exist to enable this, a brief summary of these approaches is provided below.

Fine Tuning

Fine-tuning is the process of further training a pre-trained model on task- or domain-specific data by updating its existing parameters or, in some approaches, by introducing additional trainable parameters.[Mos23] By further training the model on a domain-specific dataset, for example technical data on extraterrestrial habitats, a base model such as GPT-5 can acquire contextual knowledge of the application domain and generate its responses accordingly.

However, a significant downside of this approach is the increased risk of model hallucinations. Hallucinations refer to the phenomenon in which a model generates incorrect or fabricated information in response to a given prompt.[GYA⁺24] Such behavior can pose a significant risk in safety-critical environments.

Retrieval Augmented Generation

Retrieval-Augmented Generation (RAG) introduces relevant external information to the language model without modifying the model's internal parameters.

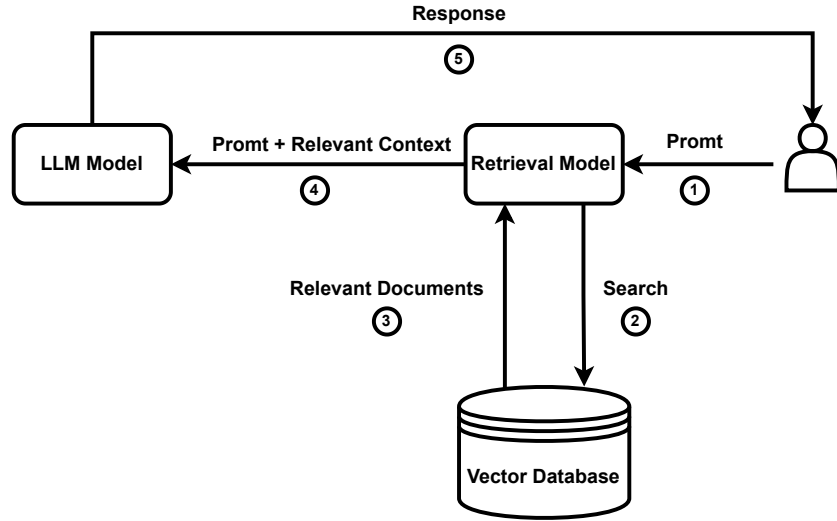


Figure 1: High-level RAG workflow that injects retrieved documents into the prompt instead of altering model weights.

As shown in Figure 1, the RAG workflow begins with a user prompt; however, instead of passing the prompt directly to the LLM, a similarity search is first performed over context-relevant information stored as embeddings in a vector database. The retrieved context is then provided together with the prompt to the LLM for grounded response generation.[BMR⁺20] Recent studies have shown this to be a more reliable method for context retrieval compared to fine-tuning.[OBME24] However the risk of hallucinations even with RAG is still a non zero possibility. However with good

Tool Use / Function Calling

Agent Architectures

2.2 Speech-to-Text, LLM and Text-to-Speech

A large body of systems research still follows the classic speech-to-text pipeline: incoming speech is transcribed, processed by an LLM, and subsequently rendered back to audio through text-to-speech engines. This architecture remains attractive because the intermediate textual representation makes it straightforward to integrate retrieval, grounding, or enterprise knowledge bases, even though the multi-stage pipeline can add noticeable latency.

2.3 Speech-to-Speech LLMs

Recent work on speech-native LLMs, popularized by OpenAI's realtime models, reduces the round-trip delay by operating directly on streaming audio chunks. The model begins producing

speech tokens almost immediately, providing near-realtime turn-taking during conversations. These systems trade some reasoning depth compared with larger text-first models such as GPT-4o or GPT-5, and incorporating retrieval-augmented generation is still an active research topic, but they mark the state of the art for responsive spoken interactions.

2.4 Vision-Language(-Action) Models

Vision-language(-action) models extend LLM capabilities to video feeds, sensor data, and embodied decision making. They process multimodal streams to ground language in the physical environment and to suggest or execute actions. Although still nascent, this strand of work is highly relevant for human-machine interaction scenarios that require situational awareness or co-robotic collaboration.

3 Domain Case Studies

In this section I will be discussing select papers that are relevant to this seminar topic. Exactly which papers are to be determined. Potential research direction is given below.

3.1 Aerospace and Remote Science

Aerospace, Deep Sea wayages etc.

Psychological and Cognitive Impacts

I am sure in long isolated voyages the use of a somewhat human-feeling interface is a lot better from a psychological aspect rather than a GUI. I have found a paper regarding this. It is certainly something I would like to involve in this paper.

3.2 Emergency and Environmental Response

3.3 Industrial Operations

4 Comparative Analysis and Metrics

I will be comparing the methods I have found in the Domain Case Studies section.

5 Safety, Trust, and Deployment Constraints

Safety is a huge topic regarding the use of AI in any system that may endanger lives. And hallucinations are a huge issue. I will be looking into more papers in this direction. At work we had huge problems with our customer because the realtime speech models hallucinate on the very extreme end. Several times they were making things up regarding our customers' data. Ours is a CRM system; in the worst case our customers would have to deal with angry clients, and such a thing can result in deaths in aerospace etc.

specieal working environments, local environments what kind of LLM technologies be used in special environments no internet, no electricity etc etc.

6 Conclusion

This section will be my conclusion.

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